FAST School of Computing

National University of Computer & Emerging Sciences Lahore Campus

Quiz 2; Sub.: Engineering Economics (MG2002); Fall 2025; Class: BCS-3A; Max. Time: 20 minutes; M/Marks: 15

Name of the Candidate: . .... Roll Number:

Note: Q 10 and Q 11 carry 3 marks each while rest carry 1.

Select the best choice.

1. Time value of money indicates that
   A. A unit of money obtained today is worth more than a unit of money obtained in future
   B. A unit of money obtained today is worth less than a unit of money obtained in future
   C. There is no difference in the value of money obtained today and tomorrow
   D. None of the above
2. How are factor values arranged in the tables for interest rates and time periods?
   A. Alphabetically
   B. Numerically
   C. By interest rate and number of periods
   D. By factor name and interest rates
3. What is the cash flow diagram for finding F, given P?
   A. (a) F given 0, 1, 2, (b) P given 0, 1, 2
   B. (a) F given 1, 2, n, (b) P given 0, 1, 2
   C. (a) F given 0, 1, 2, (b) P given 1, 2, n
   D. (a) F given 0, 1, 2, (b) P given 1, 2, n
4. What is the equivalent investment required in 3 years?
   A. 200(1.3310)
   B. 200(0.6830)
   C. 200(1.10)
   D. 200(0.9090)
5. What is the most fundamental factor in engineering economy?
   A. Uniform series factor
   B. Arithmetic gradient factor
   C. Compound amount factor
   D. Geometric gradient factor
6. What is true for (F/P, *i*%, *n*)?
   A. SPPWF
   B. SPCAF
   C. P/F factor
   D. None of the above
7. The amount of money that Diamond Systems can spend now for improving productivity in lieu of spending $30,000 three years from now at an interest rate of 12% per year is closest to:
   A. $15,700
   B. $17,800
   C. $19,300
   D. $21,350
8. Present value tables for annuity cannot be straight away applied to varied stream of cash flows. (True / false)
9. Interest rate and rate of return are the same and refer to the investor. (True / false)

10. Find the numerical value of the factors (A/P ,13%,15) using the formula. Show complete working.

$(A/P, i\%, n) = [i(1+i)^n / ((1+i)^n - 1]$

$= [0.13(1+0.13)^{15} / (1+0.13)^{15} - 1]$

$= [(0.13)(1.13)^{15} / (1.13)^{15} - 1]$

$= \frac{0.13 \times 5.4736}{5.4736 - 1}$

$= 0.7116 / 4.4736$

$= 0.1591$

$= 15.91\%$

11. How much can ABC, Inc., afford to spend now on an energy management system if the software will save the company $21,300 per year for the next 5 years? Use an interest rate of 10% per year. Show complete working.

A = $21,300

Interest Rate = 10%

Time period = 5 years

$P = A \times [(1+i)^n - 1] / [i(1+i)^n]$

$P = \frac{A[((1+i)^n - 1]}{i(1+i)^n}$

$P = \frac{21300[(1+0.10)^5 - 1]}{0.10(1+0.10)^5}$

$P = \frac{21300[(1.1)^5 - 1]}{0.1(1.1)^5}$

$P = \frac{(21300)(1.6105 - 1)}{(0.1)(1.6105)}$

$P = \frac{(21300)(0.6105)}{(0.1)(1.6105)}$

$P = 21300 \times 3.7908$

$P = \$80,744.04$

(Must confirm all the answers, they might be wrong)

ABC Inc. can afford around $80,744.04 to spend on the energy management system.